

Task 04: Graph Neural Network for 2D Neutrino Interaction Position Reconstruction

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1 Introduction

The goal of this task was to reconstruct the two-dimensional neutrino interaction position from measured Cherenkov light. Each event consists of a variable-length set of detected photon hits, where each hit contains the detection time and detector position, and the target is the true interaction coordinate $(x_{\text{true}}, y_{\text{true}})$.

2 Method

The input hit features and target coordinates were standardized using the mean and standard deviation computed on the training set. The same normalization constants were applied to the validation and test sets, and final predictions were transformed back to the original coordinate scale in m.

The reconstruction error was evaluated using the Euclidean distance

$$d_{\text{err}} = \sqrt{(x_{\text{pred}} - x_{\text{true}})^2 + (y_{\text{pred}} - y_{\text{true}})^2}.$$

Each event was represented as a graph whose nodes are photon hits with normalized time and detector position features. A Dynamic Edge Convolution GNN with two DynamicEdgeConv layers, global mean pooling, and a final MLP was used to predict the normalized interaction coordinates.

The model was trained with mean squared error loss and the Adam optimizer for 30 epochs. The best checkpoint was selected using the validation loss, with the selected checkpoint at epoch 15 and early stopping patience set to 6.

3 Results

Figure 1 shows the training and validation loss curves. The validation loss decreased during training, and the best checkpoint was selected at epoch 15. The test loss shown

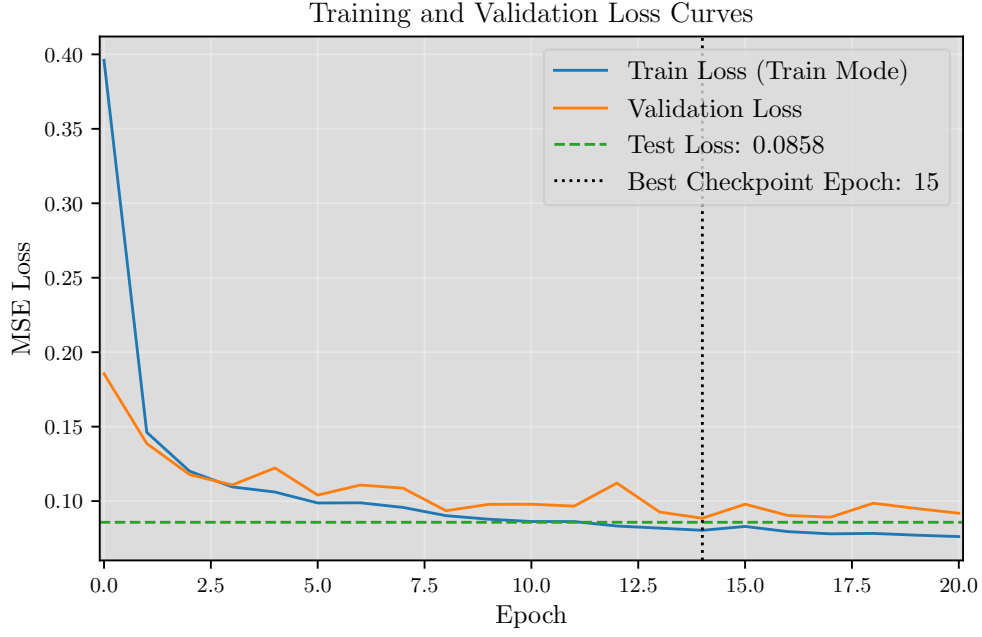


Figure 1: Training and validation loss curves. The dashed horizontal line indicates the test loss, and the vertical dotted line indicates the selected best checkpoint epoch.

in the plot is in normalized coordinate space, since the training objective was computed on normalized labels.

Figure 2 compares the predicted and true interaction coordinates on the test set after transforming the coordinates back to the original scale. The predictions follow the diagonal trend, indicating that the model learned the main structure of the reconstruction task. The distribution of reconstruction errors is shown in Figure 3. Most events have moderate errors, while the tail of the distribution corresponds to events with larger position reconstruction error.

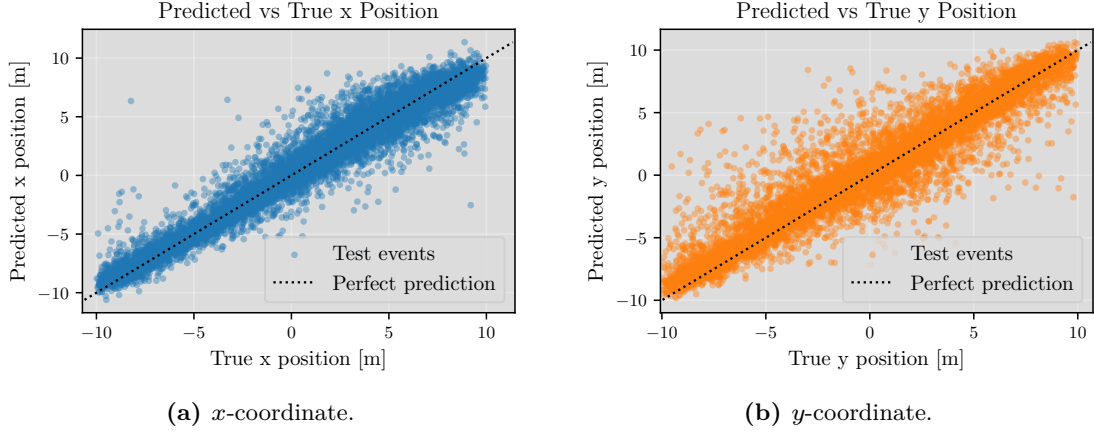


Figure 2: Predicted and true interaction coordinates on the test set after transforming the labels back to the original coordinate scale in m. The diagonal line indicates perfect reconstruction.



Figure 3: Distribution of the Euclidean reconstruction error d_{err} on the test set in m. The legend reports the mean, standard deviation, SEM, and median. Mean is the average reconstruction error, standard deviation describes the spread of event-wise errors, SEM is the standard error of the mean, and median is the central value of the error distribution.

Figure 4 visualizes 300 randomly selected test events. True and predicted positions are connected by lines whose color represents d_{err} , while the line width also varies slightly with the error. The detector positions are shown as well and were determined by collecting the unique detector hit positions from the data. This plot suggests that reconstruction quality may depend on spatial position, since errors appeared larger farther away from the detector region.

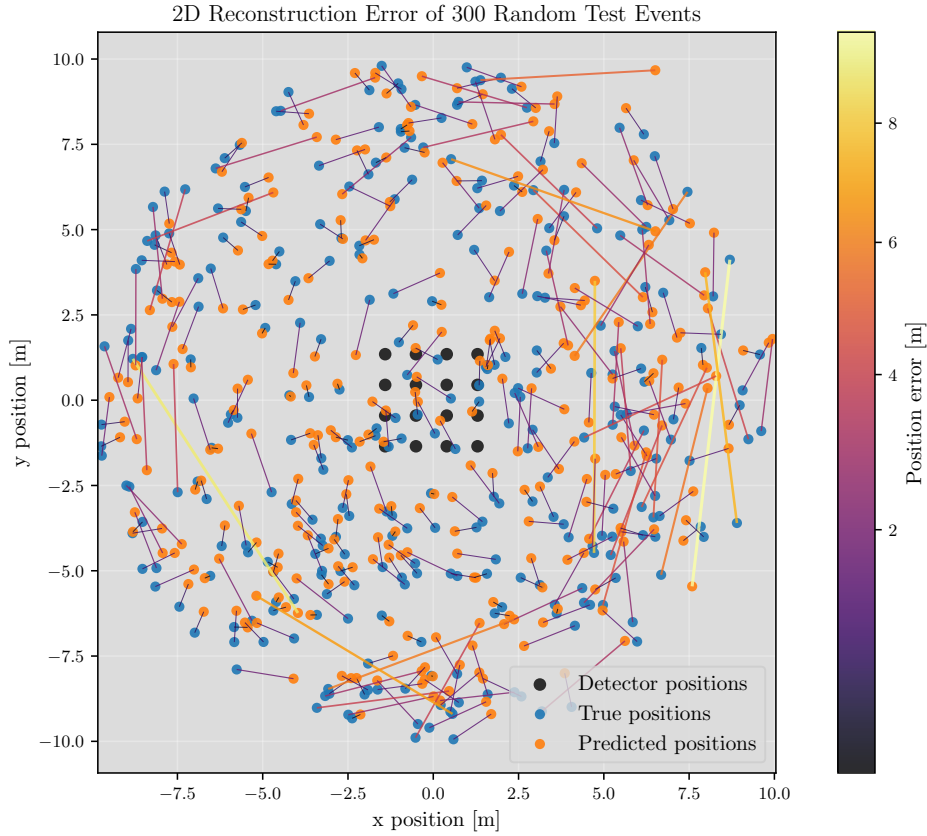


Figure 4: Two-dimensional reconstruction visualization for 300 randomly selected test events. True and predicted positions are connected by lines colored according to the Euclidean reconstruction error d_{err} in m, as shown by the colorbar. The line width also varies slightly with the error, and detector positions are shown.

The spatial dependence of the reconstruction quality is studied more directly in Figure 5. Test events were binned according to their true interaction position $(x_{\text{true}}, y_{\text{true}})$. For each bin, the color represents the mean Euclidean reconstruction error, while the number inside the bin gives the standard deviation of the error values in that bin. The heatmap confirms that the reconstruction quality varies with spatial position, with larger errors in some outer regions.

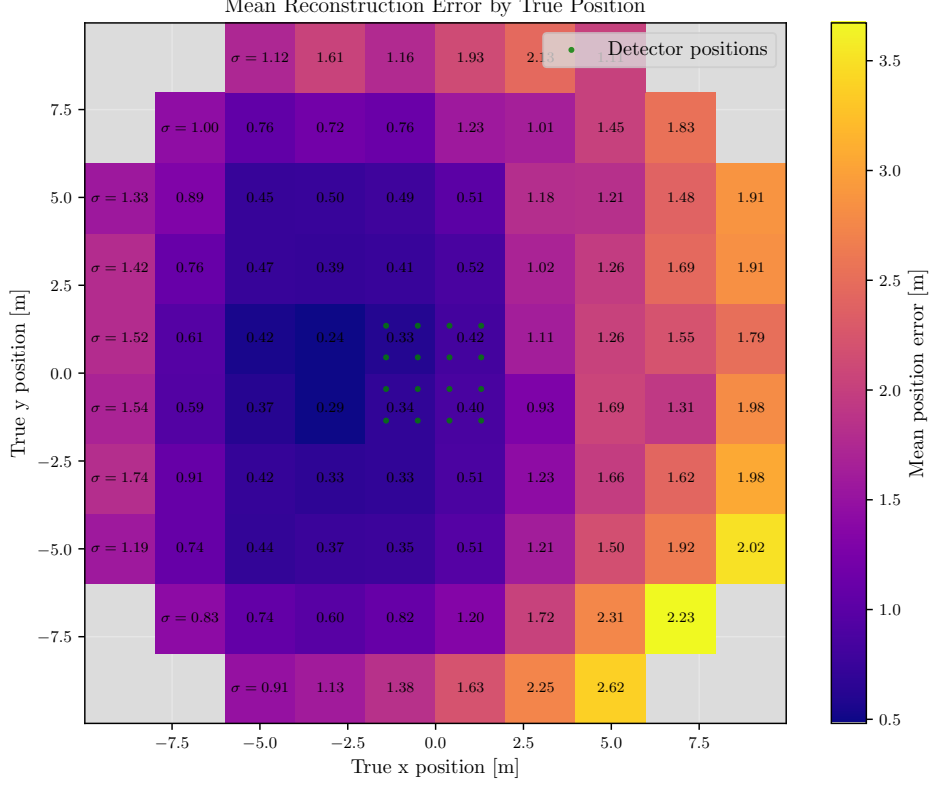


Figure 5: Spatial dependence of reconstruction quality on the test set. Events are binned by true interaction position, the color gives the mean Euclidean reconstruction error in m, and the number inside each bin gives the standard deviation of the error values in that bin. Detector positions are marked in the plot.

4 Conclusion

The Dynamic Edge Convolution GNN was able to learn the reconstruction task and produced reasonable predictions on the test set. The spatial plots show that the reconstruction performance is not uniform over the detector plane, with larger errors in some outer regions.

As possible extensions, the GNN could be compared to baseline approaches such as a

simple fully connected network using fixed summary features, a point-cloud model, or a model based on handcrafted event-level quantities. Future work could also include fixing random seeds and documenting the exact training configuration more carefully for reproducibility. All code used for preprocessing, training, evaluation, and plotting is available in the accompanying GitHub repository.